

FERNANDINA BEACH, FL, USA

WMO: 997347

Lat: 30.6753N

Lon: 81.4651W

Elev: 23

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 34.7	(c) 38.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 23.9	(k) 54.1	(l) 20.9	(m) 51.6	(n) 11.0	(o) 330	(p) 0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.5	(c) 92.7	(d) N/A	(e) 90.4	(f) N/A	(g) 88.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.8	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.1	(b) 17.4	(c) 15.4	(e) 30.0	(f) 95.4	(g) 4.3	(h) 3.2	(i) 26.9	(j) 97.7	(k) 24.4	(l) 99.6	(m) 21.9	(n) 101.4	(o) 18.8	(p) 103.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	53.9	57.9	62.0	68.8	75.5	80.7	82.4	82.9	80.3	73.2	63.0	59.3
	DBStd	11.70	8.76	8.13	7.08	5.35	4.32	2.82	2.44	2.72	3.63	6.28	6.96	8.15
	HDD50	110	61	21	7	0	0	0	0	0	0	0	4	17
	HDD65	1076	353	220	140	26	0	0	0	0	0	13	114	210
	CDD50	7431	183	243	379	564	790	922	1004	1020	909	718	394	305
	CDD65	2921	10	22	46	140	325	472	539	555	459	266	54	33
	CDH74	25407	9	49	98	486	2027	4497	5739	6146	4498	1678	122	58
CDH80	7524	0	1	4	51	402	1331	1955	2211	1268	293	5	3	
Wind	WSAvg	7.0	7.4	7.0	7.6	7.6	7.0	6.7	6.4	5.9	6.5	6.9	7.5	7.4
Precipitation	PrecAvg	51.8	3.4	3.6	3.9	2.7	3.0	5.6	6.0	6.2	7.6	4.3	2.3	2.9
	PrecMax	75.4	9.4	10.1	10.0	8.9	8.4	11.1	13.3	20.1	23.8	13.8	5.4	12.3
	PrecMin	32.2	0.2	0.7	0.1	0.2	0.1	1.3	1.4	2.2	0.7	0.2	0.0	0.0
	PrecStd	9.8	2.2	2.2	2.5	1.8	2.1	2.6	2.7	3.5	4.8	3.6	1.6	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	78.5	80.6	84.6	92.3	93.8	94.9	95.2	94.8	89.4	80.6	78.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	71.7	75.2	76.7	81.4	86.9	90.7	91.4	92.4	89.5	85.5	77.4	75.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	68.9	71.6	74.4	78.9	84.0	88.4	89.4	89.8	87.2	82.9	75.2	72.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	66.5	68.8	71.7	76.9	81.8	86.4	87.4	87.9	85.7	81.1	73.0	70.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	13.3	13.3	13.0	12.3	10.8	11.0	11.0	10.5	8.7	10.2	11.7	12.6
		MCDBR	14.6	15.4	14.8	14.3	14.3	14.1	13.5	13.1	10.9	10.6	12.8	13.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.338	0.348	0.373	0.399	0.440	0.486	0.510	0.495	0.429	0.401	0.360	0.348
	taud		2.533	2.510	2.455	2.372	2.291	2.221	2.160	2.209	2.389	2.431	2.498	2.532
	Ebn at Noon		283	291	289	284	271	258	251	254	268	268	272	274
	Edh at Noon		28	31	35	39	43	45	48	45	36	33	28	26
All-Sky Solar Radiation	RadAvg		974	1170	1537	1890	2037	1922	1918	1770	1507	1325	1066	873
	RadStd		67	134	143	115	118	128	121	110	127	117	62	74

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	+0.81	+1.37	N/A	N/A	N/A	N/A	-29	-122	+288	+205

Nomenclature: See separate page